ROAD TRAFFIC ACCIDENT ANALYSIS AND PREDICTIVE MODELING

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Big Data and Data Mining

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Abstract

Road traffic accidents are a significant public health concern, contributing to injuries, fatalities, and substantial economic costs worldwide. This study analyzes road traffic accident data from the United Kingdom to identify factors influencing accident severity. Key variables examined include road types, weather conditions, time of day, and accident patterns involving motorcyclists and pedestrians. The analysis reveals that most accidents are slight, but serious and fatal accidents account for a considerable portion of the total, emphasizing the need for targeted interventions. Predictive models and social network analysis further highlight critical areas for improvement, such as dual carriageways and adverse weather conditions. The findings provide valuable insights for developing data-driven safety measures and more efficient traffic management strategies to reduce accident frequency and severity, ultimately enhancing road safety.

1 Introduction and Background

Road traffic accidents (RTAs) are a pervasive global issue, representing one of the most significant challenges to public health, safety, and sustainable development in the 21st century. The World Health Organization (WHO) estimates that approximately 1.3 million people lose their lives annually due to road traffic collisions, with an additional 20–50 million individuals suffering non-fatal injuries, many of which result in long-term disabilities (WHO, 2024). This staggering death toll surpasses fatalities from some of the world's most severe diseases, positioning RTAs as the leading cause of death among individuals aged 15–29 (WHO, 2024). Despite advances in vehicle technology and global road safety campaigns, the burden of RTAs remains disproportionately high, especially in low- and middle-income countries, where over 90% of traffic-related deaths occur despite accounting for only 60% of the world's vehicles (Ahmed et al., 2023).

The socioeconomic impact of RTAs cannot be overstated. Beyond the immeasurable personal tragedies, the economic cost is estimated to account for 3% of a country's gross domestic product (GDP) in many regions, encompassing direct medical expenses, lost productivity, legal proceedings, and the repair or replacement of damaged infrastructure (Kumar, 2024). In developed countries like the United Kingdom, road traffic collisions persist as a leading cause of injury and death. Data from Thompsons Solicitors (2023) reveal a concerning prevalence of severe and fatal accidents, despite stringent road safety regulations, advancements in vehicle safety features, and public awareness campaigns. This persistence underscores the multifaceted nature of RTAs, which result from an interplay of human error, vehicle condition, infrastructure deficiencies, and environmental factors (Hammad et al., 2019).

Critically, addressing RTAs requires a paradigm shift in understanding and intervention. While traditional approaches to road safety have emphasized reactive measures such as post-accident response and passive vehicle safety systems, emerging research highlights the need for a proactive, data-driven approach. This includes leveraging real-world accident data to identify and analyze patterns, contributing factors, and at-risk populations. For instance, studies have consistently demonstrated the role of speed, alcohol impairment, fatigue, and distracted driving in exacerbating accident severity (Fernandes et al., 2010). However, broader systemic issues, such as poor urban planning, lack of pedestrian infrastructure, and inequitable access to safe transportation, exacerbate these risks, particularly in underserved communities (Litman, 2024). This study aims to contribute to this critical area by undertaking an in-depth analysis of road traffic accident data, focusing on identifying key determinants of accident severity. By examining patterns associated with environmental conditions (e.g., weather, lighting, and time of day), road characteristics (e.g., road type and geometry), and human factors, the research seeks to uncover actionable insights for mitigating accident risks. These insights are particularly important in light of climate change, urbanization, and increasing traffic volumes, which collectively amplify the complexity of road safety management.

Moreover, this research moves beyond traditional descriptive approaches by integrating predictive modeling. Using advanced machine learning techniques, predictive models will be developed to forecast accident outcomes based on diverse variables. Predictive analytics not only provide a deeper understanding of contributory factors but also enable authorities to implement targeted interventions, such as optimized traffic signal timing, road design improvements, and real-time driver alerts. This forward-looking approach aligns with global commitments to sustainable development, particularly Sustainable Development Goal (SDG) 3.6, which aims to halve global road traffic deaths and injuries by 2030 (United Nations, 2015).

The significance of this research lies not only in its potential to reduce RTAs but also in its contribution to the broader discourse on transportation safety. By adopting a holistic and interdisciplinary lens, this study aspires to bridge existing knowledge gaps and propose evidence-based solutions that prioritize equity, sustainability, and resilience. Ultimately, the findings are expected to support policymakers, urban planners, and public health officials in designing safer, more adaptive transportation

systems that reflect the complex realities of modern mobility.

2 Methods

The following methods will be used in the analysis of this report:

i **Descriptive Statistics** Descriptive statistics summarize and describe the main features of a dataset using measures such as mean, median, mode, standard deviation, and range. These statistics provide a quick understanding of the data's distribution, central tendency, and variability.

ii **Trend Analysis** Trend analysis identifies patterns or directions in data over time, such as upward or downward movements. It is commonly used to detect long-term tendencies and predict future behavior based on historical data.

iii **Association Rule Mining** Association rule mining uncovers relationships or patterns among variables in large datasets. For example, it identifies rules like "If item A is purchased, item B is likely to be purchased" and is widely used in market basket analysis.

iv **Logistic Regression** Logistic regression is a supervised learning method used for binary or multi-class classification tasks. It models the relationship between a dependent categorical variable and one or more independent variables, providing probabilities for each outcome.

v **Spatial Analysis** Spatial analysis examines data in a geographical or spatial context to identify patterns, relationships, and trends. It is used in fields like GIS, urban planning, and environmental science for analyzing location-based data.

vi **Time Series Forecasting** Time series forecasting predicts future values based on past observations of a time-ordered dataset. Techniques like ARIMA, exponential smoothing, and deep learning models are commonly used in applications such as sales forecasting and weather prediction.

vii **Deep Learning Models for Sequential and Text Analysis** This category includes advanced neural network architectures like Long Short-Term Memory (LSTM) networks and Bidirectional Encoder Representations from Transformers (BERT).

- LSTM is a type of recurrent neural network particularly effective at modeling sequential data and long-term dependencies. It is applied in time series analysis, text classification, and sentiment analysis.
- BERT is a transformer-based model designed for natural language understanding tasks. It captures contextual relationships between words in text, making it useful for tasks like sentiment analysis, text classification, and question answering.

3 Analyses and Results

3.1 Data Description:

The dataset consists of detailed records of road traffic accidents and related information, organized into four main tables. The Accident table contains 36 columns and 461,352 rows. Key variables include accident severity, which classifies accidents as fatal, serious, or slight; road type, describing the type of road where accidents occurred; weather and light conditions at the time of the accident; road surface conditions, such as whether the road was dry, wet, or icy; and the date and time of the accident.

The Casualty table includes 19 columns and 600,332 rows, focusing on casualty severity, categorizing injuries as fatal, serious, or slight. It also details the type of casualty (e.g., driver, pedestrian, or passenger), age and gender of those involved, the number of car passengers, and the area type of the casualty's residence. The LSOA table, with 7 columns and 34,378 rows, provides geographic details about accident locations, including LSOA names and codes as unique identifiers, easting and northing coordinates for mapping, and the broader region name, such as the city or borough. The Vehicle table, comprising 28 columns and 849,091 rows, captures vehicle-specific information. It includes vehicle reference and accident index to link vehicles to specific accidents, the age of drivers, the type of vehicles involved, and their maneuvers at the time of the accident. Together, these tables provide a comprehensive dataset to analyze road traffic accidents, including accident characteristics, casualty details, geographical context, and vehicle information.

3.2 Initial Data Exploration and Summary Analysis:

3.2.1 Accident Severity Distribution:

The accident severity distribution as shown in Figure 1 reveals that 79.83% of all recorded accidents were classified as slight, indicating that a majority of accidents did not result in severe injuries. Serious accidents accounted for 18.78%, involving more significant injuries or damage. Fatal accidents, on the other hand, made up only 1.39% of the total accidents, which, while small, is a crucial percentage for safety improvements.

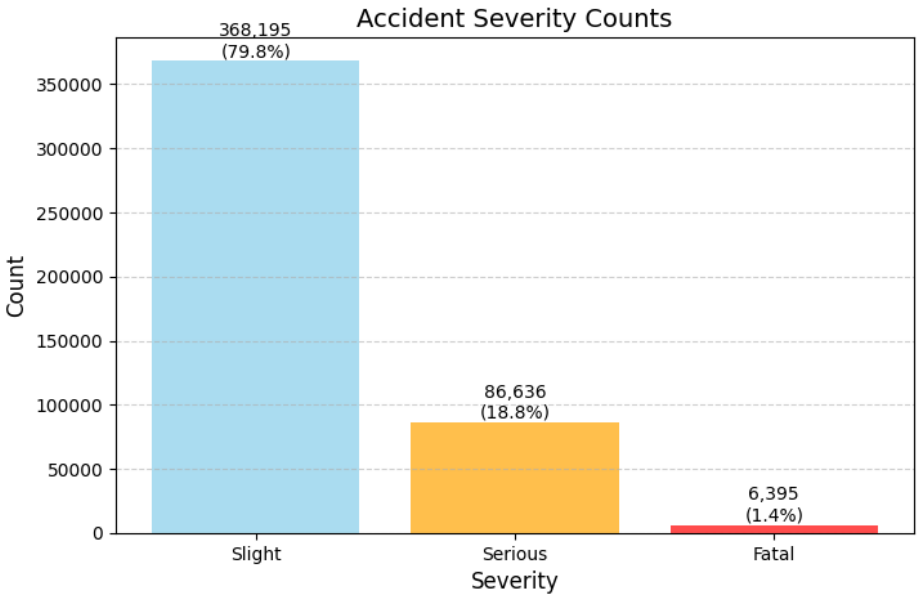

Figure 1: Accident severity counts

3.2.2 Road Types and Accident Severity:

The distribution of accident severity varies across different road types, reflecting distinct patterns of risk and outcomes as presented in Figure 2. On roundabouts, slight accidents dominate, making up 86.05% of all incidents. Serious accidents account for 13.57%, and fatal accidents are relatively rare at 0.38%. Similarly, on one-way streets, slight accidents constitute 83.66%, while serious accidents are higher at 15.77%, and fatal accidents occur at 0.5Dual carriageways present a different distribution,

with slight accidents comprising 81.39%, serious accidents reaching 16.86%, and fatal accidents rising to 1.75%. Single carriageways show an even higher proportion of serious accidents at 20.09%, though slight accidents still account for the majority at 78.45%, and fatal accidents are at 1.46%. Slip roads reflect a similar distribution, with slight accidents at 84.93%, serious accidents at 14.02%, and fatal accidents at 1.04%. For unknown road types, slight accidents dominate at 90.55%, while serious and fatal accidents occur less frequently, at 9.17% and 0.29%, respectively.

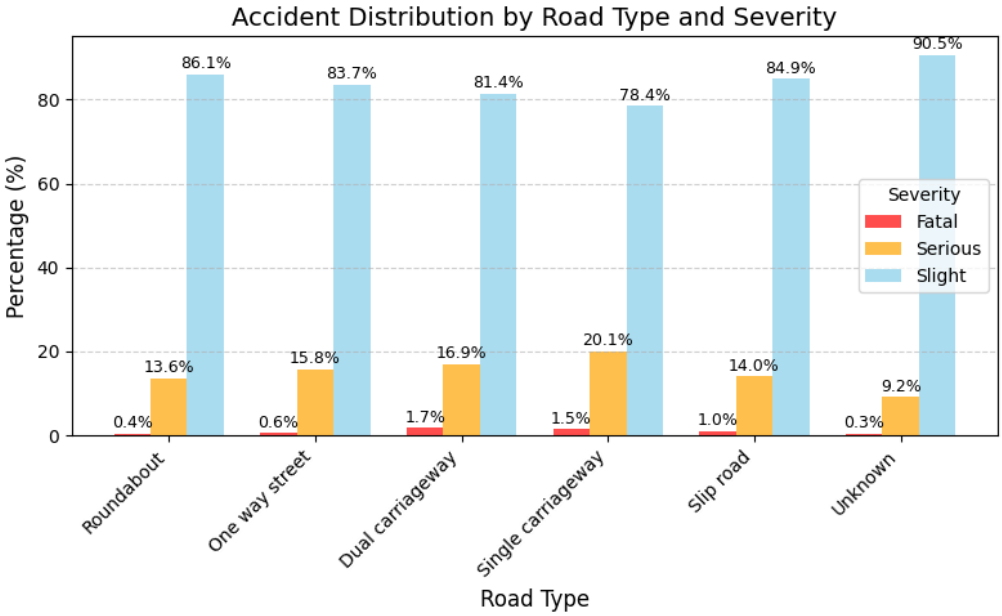

Figure 2: Accident Distribution by Road Type and Severity

3.2.3 Weather conditions and Accident Severity:

The distribution of accidents by weather condition shown in Figure 3 shows the following: In fine weather without high winds, which accounts for 79.55% of all recorded accidents, 79.36% are classified as slight accidents, 19.22% as serious accidents, and 1.42% as fatal accidents. Raining without high winds makes up 11.70% of accidents, with 80.93% slight accidents, 17.91% serious accidents, and 1.16% fatal accidents. Snowing without high winds, though less frequent, represents 0.49% of accidents, with 83.27% slight accidents, 15.49% serious accidents, and 1.24% fatal accidents. Fine weather with high winds contributes to 1.08% of accidents, with 76.12% slight accidents, 21.52% serious accidents, and 2.36% fatal accidents. Raining with high winds accounts for 1.19% of accidents, with 76.74% slight accidents, 20.92% serious accidents, and 2.34% fatal accidents. Snowing with high winds represents 0.13% of accidents, with 84.20% slight accidents, 15.31% serious accidents, and 0.49% fatal accidents.

Fog or mist, if hazardous, makes up 0.43% of accidents, with 74.78% slight accidents, 21.42% serious accidents, and 3.80% fatal accidents. Other weather conditions account for 2.37% of accidents, with 83.97% slight accidents, 15.22% serious accidents, and 0.81% fatal accidents. Finally, accidents under unknown weather conditions make up 3.06% of all accidents, with 87.06% slight accidents, 12.23% serious accidents, and 0.72% fatal accidents.

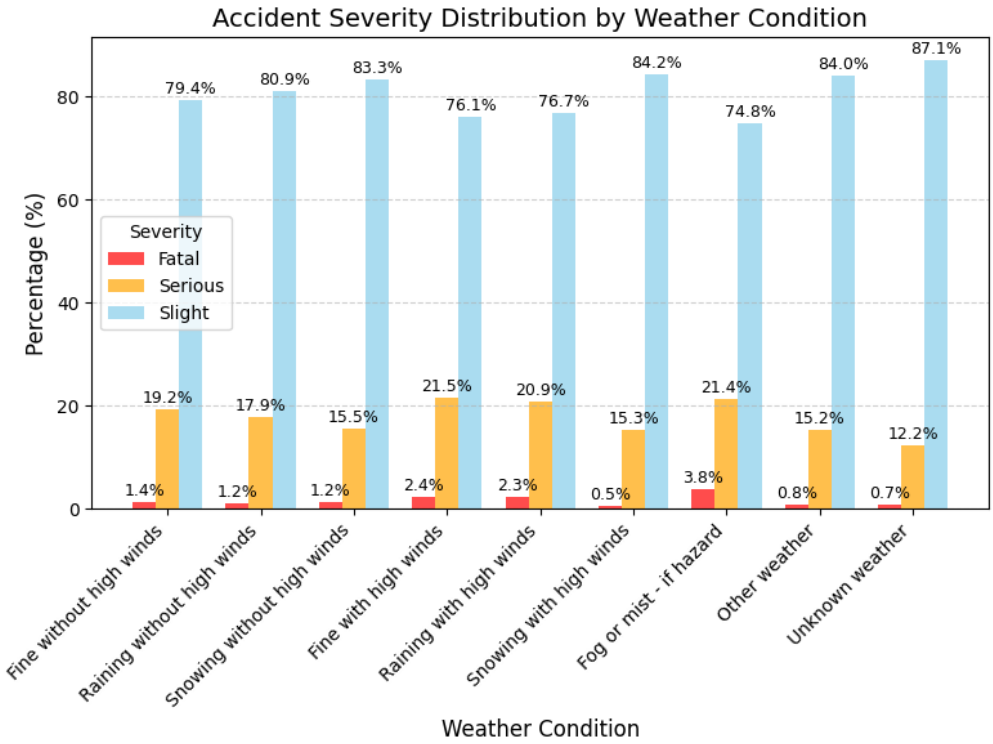

Figure 3: Accident Severity Distribution by Weather Condition

3.3 Accident Trends by Time and Demographics

3.3.1 Accidents trend by hours of the day:

The analysis of accident occurrences by hour of the day as dsiplayed in Figure 4 shows the following results: From 00:00 to 05:00, accidents are relatively low. Starting at 06:00, there is a gradual increase in accidents, with 9,362 accidents recorded. The number of accidents peaks from 08:00 to 17:00, with 31,152 accidents at 08:00 and 40,307 accidents at 17:00. The peak at 17:00 coincides with the evening rush hour. After 18:00, accident counts begin to decrease, with 33,221 accidents recorded at 18:00. The lowest counts are observed during late-night and early-morning hours when traffic is minimal.

3.3.2 Accidents trend by days of the week:

The analysis of accident occurrences by day of the week reveals the following results: Friday consistently has the highest number of accidents, with a total of 75,211 accidents. This is followed by Thursday with 70,482 accidents, and Wednesday with 69,055 accidents. The elevated accident counts on these days may be linked to increased travel and road congestion toward the end of the workweek, as people engage in work-related and social activities. In contrast, Sunday records the lowest number of accidents, with 52,028 accidents. This suggests that the start of the week sees reduced vehicular activity, potentially due to less frequent commuting and travel compared to other days. These trends highlight the importance of targeted interventions on high-accident days like Friday to improve road safety, while acknowledging the lower risks associated with Sunday. Figure 5 presents the details.

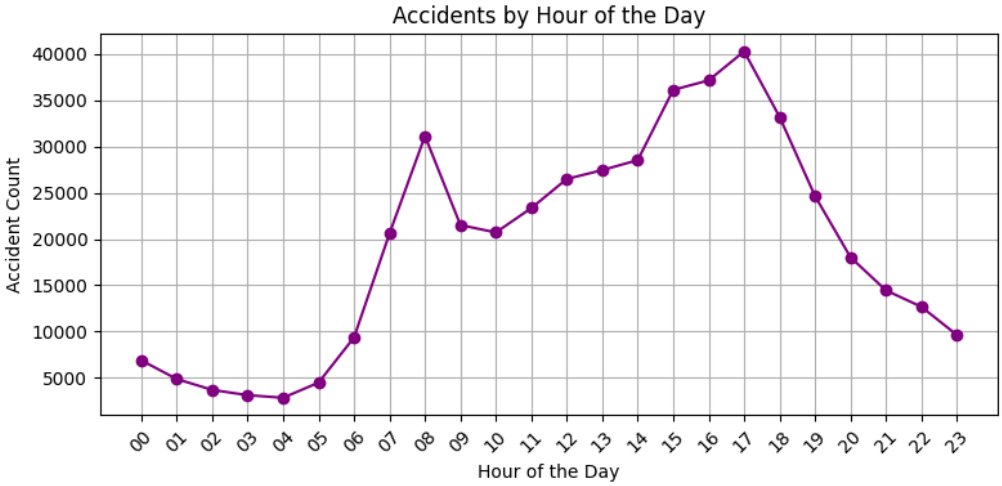

Figure 4: Accidents by Hour of the Day

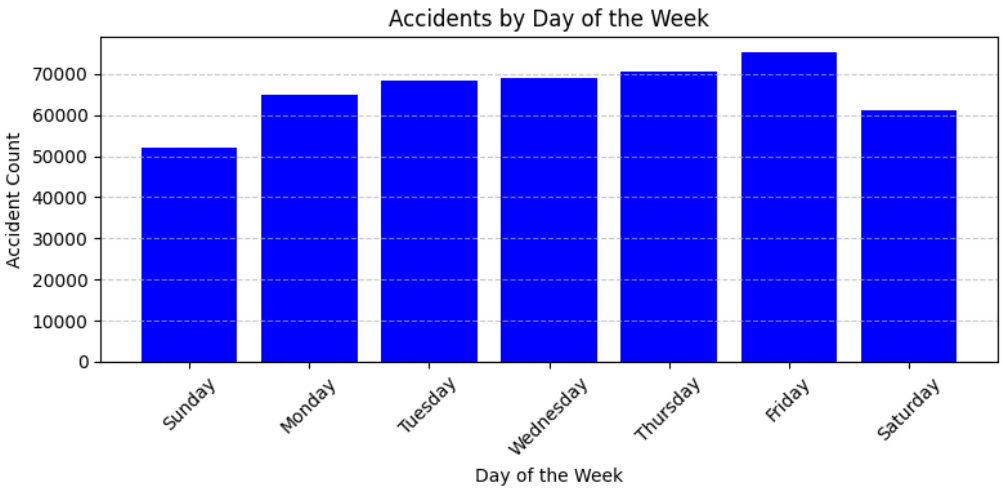

Figure 5: Accidents by Day of the Week

3.3.3 Motorcycle Accidents by Hour of the Day

Motorcycle accidents by hour of the day reveal the following patterns: For motorcycles 50cc and under as shown in Figure 6, accidents are sparse during the early morning hours but start to rise after 06:00, with the peak at 17:00 (601 accidents), coinciding with the evening rush hour. Accidents gradually decrease after 18:00 as traffic diminishes.

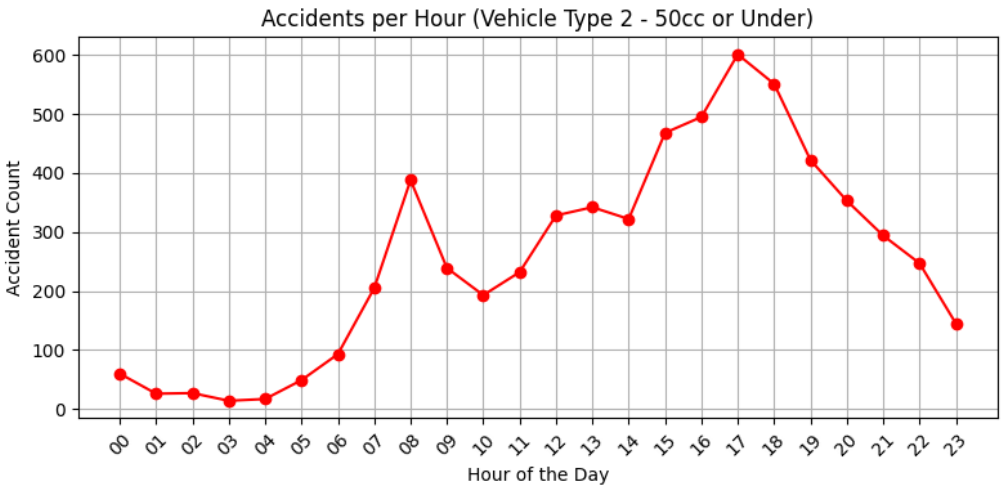

Figure 6: Accidents per Hour (Vehicle Type 2 - 50cc or Under)

Motorcycles in the 50cc to 125cc range show a similar trend as presented in Figure 7, with accidents starting to rise at 06:00 (622 accidents) and peaking at 18:00 (2,716 accidents), which aligns with the busy evening commute.

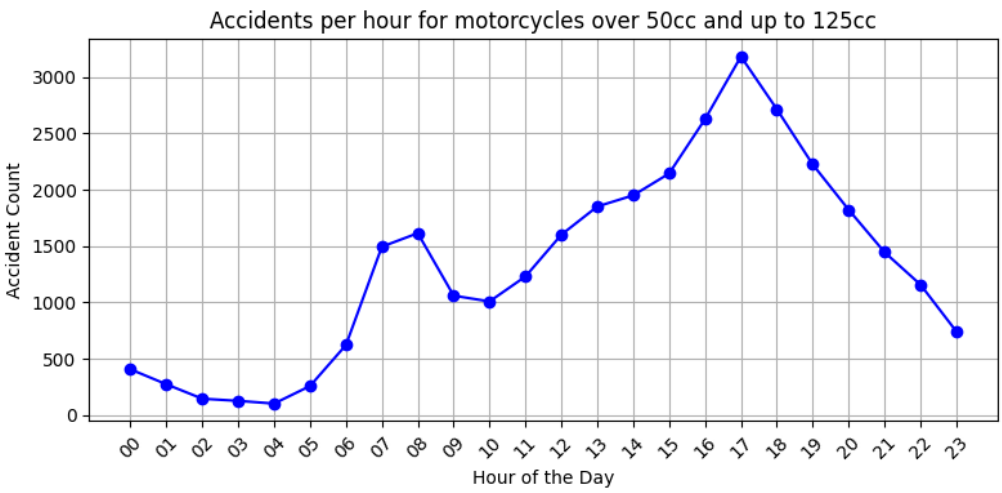

Figure 7: Accidents per hour for motorcycles over 50cc and up to 125cc)

Motorcycles between 125cc and 500cc also follow this pattern as seen in Figure 8, with peaks at 17:00 (866 accidents) and 18:00 (699 accidents), emphasizing the impact of rush hour traffic.

As shown in Figure 9, larger motorcycles (over 500cc) exhibit the highest accident count at 17:00

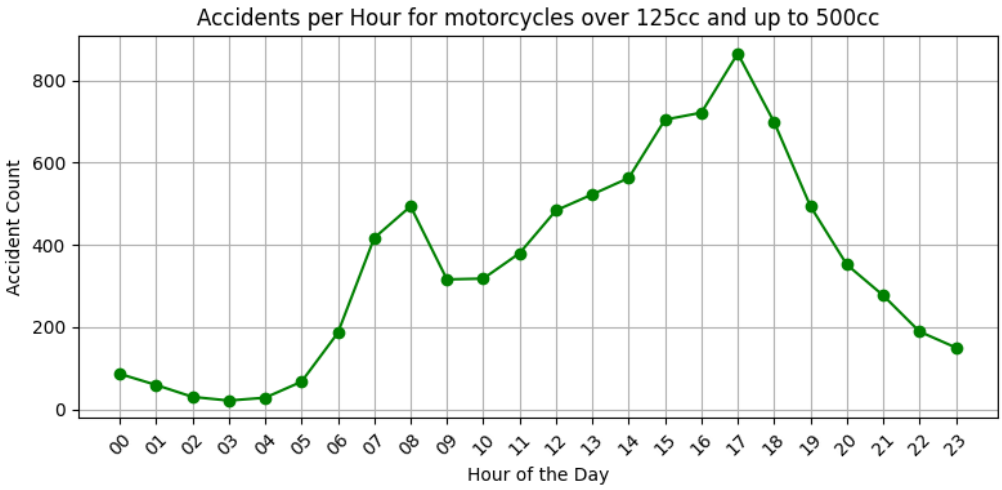

Figure 8: Accidents per Hour for motorcycles over 125cc and up to 500cc)

(2,197 accidents), followed by 18:00 (1,547 accidents). This trend is consistent across all categories, highlighting the role of heavy traffic and congestion during rush hour in increasing accident rates.

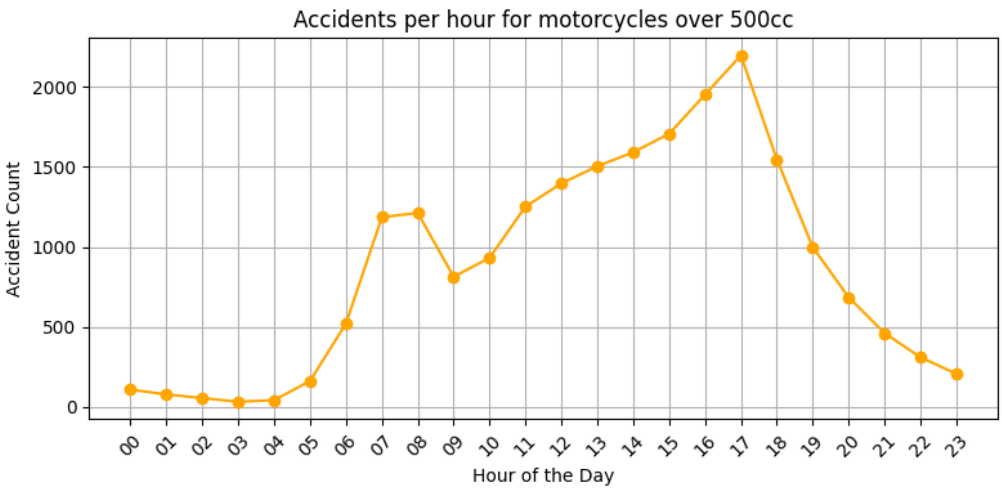

Figure 9: Accidents per hour for motorcycles over 500cc

3.3.4 Motorcycle Accidents by Day of the Week

When analyzing motorcycle accidents by day of the week, the following patterns emerge: Motorcycle accidents show clear trends across different motorcycle categories. For motorcycles 50cc and under, Friday recorded the highest number of accidents at 1,040, while Sunday had the fewest with 651 accidents. For motorcycles over 50cc and up to 125cc, Friday had the most accidents at 5,388, and Sunday the least at 3,380. For motorcycles over 125cc and up to 500cc, Thursday saw the highest number of accidents at 1,372, while Sunday recorded the fewest with 1,025 accidents. For motorcycles over 500cc, Sunday exhibited the highest number of accidents at 3,443, followed by Friday with 3,164 accidents. This suggests that smaller motorcycles (50cc to 125cc) tend to have higher accident counts on Fridays, likely due to increased road activity at the end of the workweek. In contrast, larger

motorcycles (over 500cc) see more accidents on Sundays, which may be linked to recreational riding or leisure activities. Motorcycle accidents by day of the week are presented in Figures 10-13 for all categories of motorbikes.

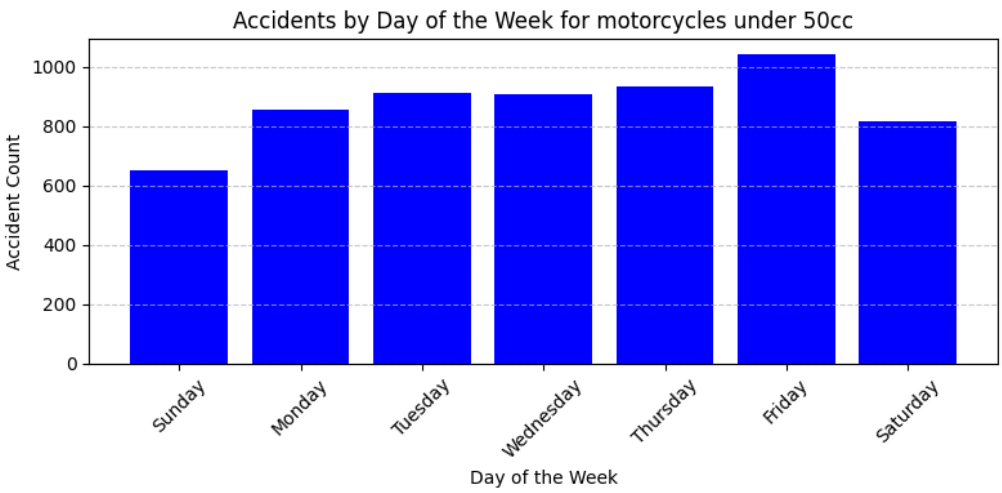

Figure 10: Accidents by Day of the Week for motorcycles under 50cc

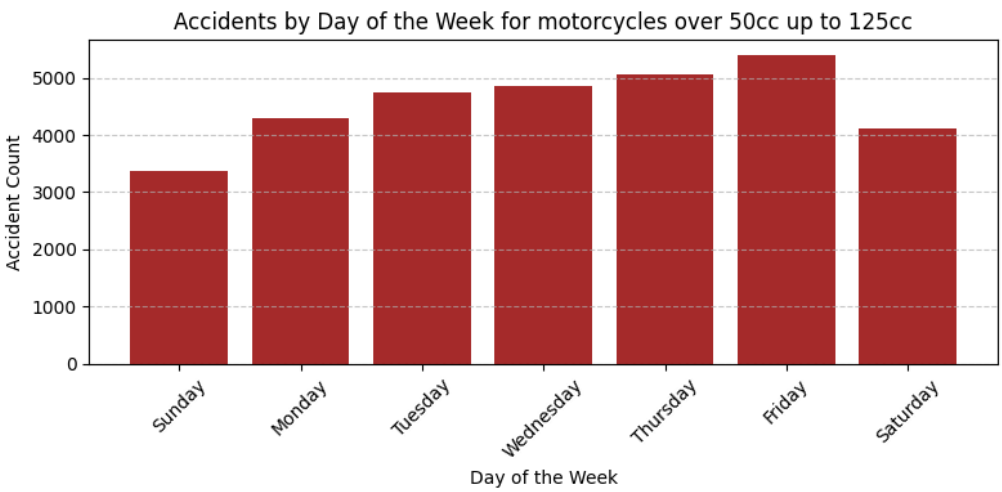

Figure 11: Accidents by Day of the Week for motorcycles over 50cc up to 125cc

3.3.5 Pedestrian Accidents by Hour of the Day

Pedestrian accidents show a clear pattern tied to peak commuting hours. The number of accidents begins to rise notably at 08:00, reaching 6,448 incidents, which coincides with the morning rush hour when pedestrians are on their way to work or school. The trend continues, with a significant peak occurring at 15:00, with 9,003 accidents, aligning with the afternoon rush hour when pedestrians are returning home or from work. These two periods—08:00-09:00 and 15:00-17:00—represent critical

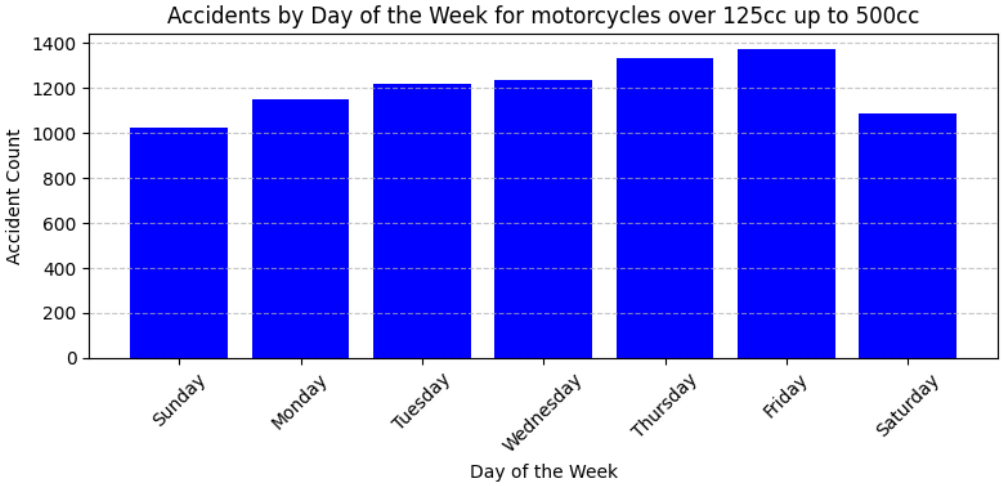

Figure 12: Accidents by Day of the Week for motorcycles over 125cc up to 500cc

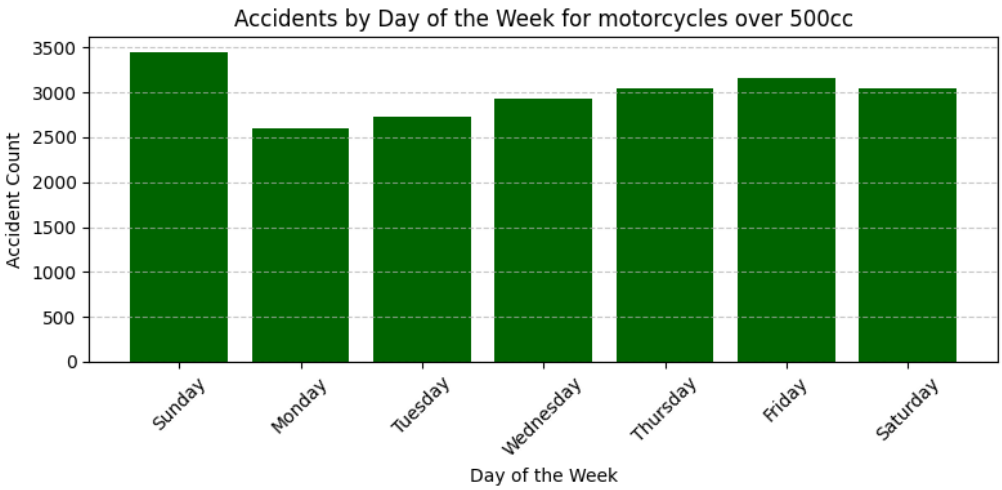

Figure 13: Accidents by Day of the Week for motorcycles over 500cc

times for pedestrian safety. In contrast, the lowest accident counts occur in the late-night and early-morning hours, particularly between 00:00-05:00, when pedestrian activity is minimal. This indicates that pedestrian accidents are predominantly higher during the day, especially when foot traffic is at its peak during commuting hours. The plot is presented in Figure 14.

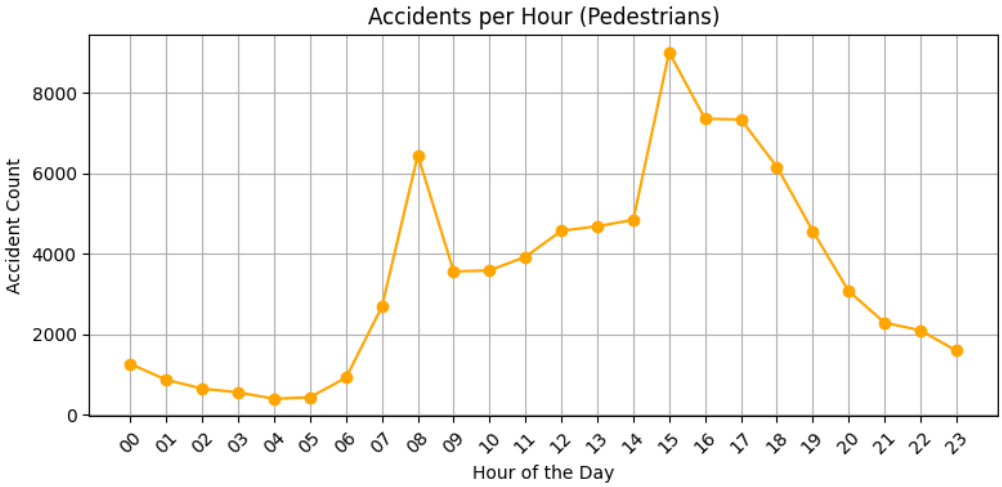

Figure 14: Accidents per Hour (Pedestrians)

3.3.6 Pedestrian Accidents by Day of the Week

Pedestrian accidents (Figure 15) reveal distinct patterns by day of the week. Friday recorded the highest accident count at 14,168 accidents, followed by Thursday with 12,985 accidents and Wednesday with 12,780 accidents. These trends indicate that the end of the workweek sees the most pedestrian accidents, likely due to increased commuting and evening social activities. Saturday had 10,729 accidents, which, while high, was lower than Friday, potentially reflecting reduced commuter traffic and more localized leisure activities. Interestingly, Sunday recorded the fewest pedestrian accidents at 7,511, suggesting decreased pedestrian activity or fewer events on this day. Monday, with 11,946 accidents, marked a consistent return to weekday patterns, characterized by moderate pedestrian activity levels.

3.4 Analysis of Accident Severity Using the Apriori Algorithm

The Apriori algorithm was utilized to investigate the relationships between accident severity and selected variables, including light conditions, weather conditions, road type, and road surface conditions. A dataset of 10,000 records was processed, with categorical variables mapped to meaningful categories to enhance interpretability. One-hot encoding was employed to transform these categorical variables into binary features, enabling the identification of frequent itemsets and association rules. The analysis yielded 178 significant association rules with varying levels of support, confidence, and lift values.

3.4.1 Impact of Light Conditions on Accident Severity

The analysis revealed strong associations between light conditions and accident severity. For instance, the rule linking *Daylight* with *Severity_Slight* showed robust support and confidence values, indicating

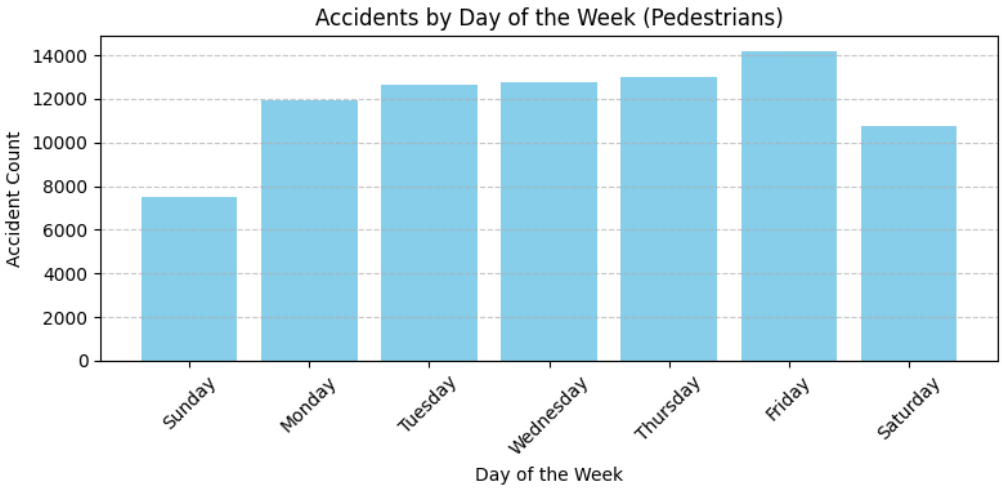

Figure 15: Accidents by Day of the Week (Pedestrians)

that slight accidents predominantly occur during daylight. This insight underscores the prevalence of lower-severity accidents in well-lit conditions.

3.4.2 Impact of Weather Conditions on Accident Severity

Weather conditions also demonstrated significant associations with accident severity. The rule associating *Severity_Serious* with *Fine weather without wind* had a support of 10.71% and a confidence of 82.38%, suggesting that serious accidents are more likely to occur in such weather. Additionally, the findings emphasized the need to consider weather variations in traffic safety measures.

3.4.3 Impact of Road Type on Accident Severity

Patterns between road types and accident severities were uncovered through the analysis. For example, the rule linking *Dual Carriageway* with *Severity_Serious* exhibited a confidence of 87.16% and a lift of 1.007, indicating a slight positive association. These findings suggest that certain road types, such as dual carriageways, are more prone to serious accidents. Implementing stricter speed limits and enhancing road safety measures on these road types could mitigate accident severity.

3.4.4 Impact of Road Surface Conditions on Accident Severity

Road surface conditions were closely associated with accident severity. The rule *Road_Dry =¿ Severity_Slight* had a high support of 61.27% and a confidence of 86.02%, highlighting the frequent occurrence of slight accidents on dry roads. This indicates that even under favorable road surface conditions, slight accidents remain common, necessitating proactive measures to address potential risk factors. Figures 16, 17, & 18 provide insights on the reles.

Visualization of the top rules highlighted the dominant role of daylight conditions in influencing accident severity.

The Support vs. Lift plot (Figure 19) shows that most points fall within a lift range of 0 to 1.5, with support values ranging from 0.1 to 0.7. This suggests that the majority of association rules are based on moderately frequent patterns (support values between 0.1 and 0.7). These associations are not extremely rare, yet their lift values indicate that the strength of the relationships varies from

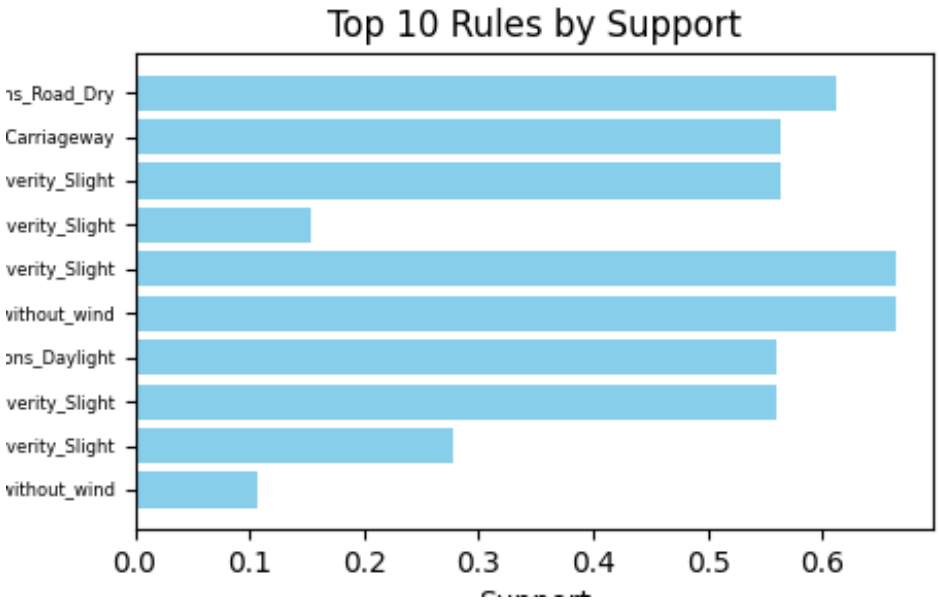

Figure 16: Top 10 Rules by Support

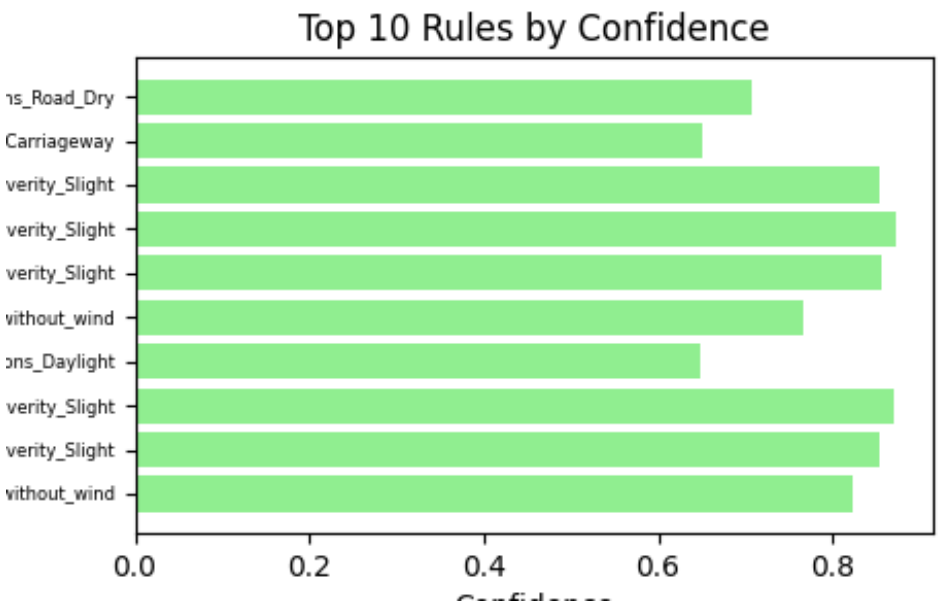

Figure 17: Top 10 Rules by Confidence

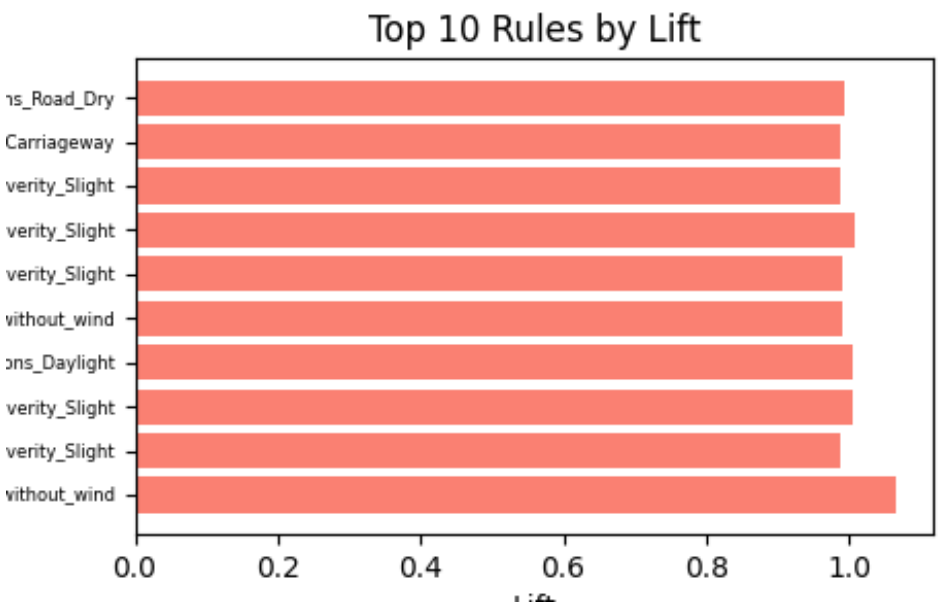

Figure 18: Top 10 Rules by Lift

weak to moderate. The clustering of points within this range implies that while many associations are relatively common, their impact on accident severity is generally modest. A few outliers with higher lift values point to some stronger relationships, which may warrant further investigation.

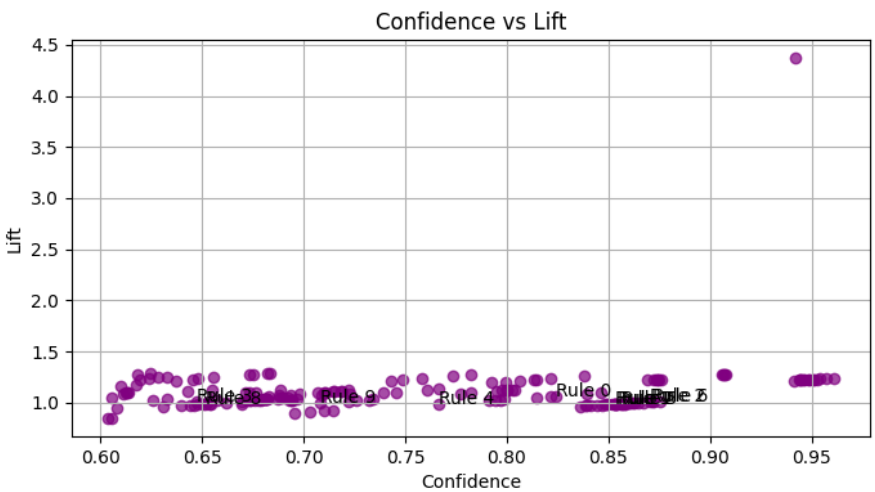

Figure 19: Confidence vs Lift

In the Confidence vs. Lift plot (Figure 20), most points fall within a confidence range of 0.6 to 1 and a lift range from 0 to 1.5. This indicates that the majority of the association rules have relatively high confidence, suggesting a strong likelihood that the occurrence of one event (such as a specific road condition) is associated with another (such as a particular accident severity). However, the lift values remaining below 1.5 indicate that these associations are not exceptionally strong, meaning the relationships identified are moderate rather than highly significant. The concentration of points in this range suggests that while the associations are reliable, they may not always represent very impactful

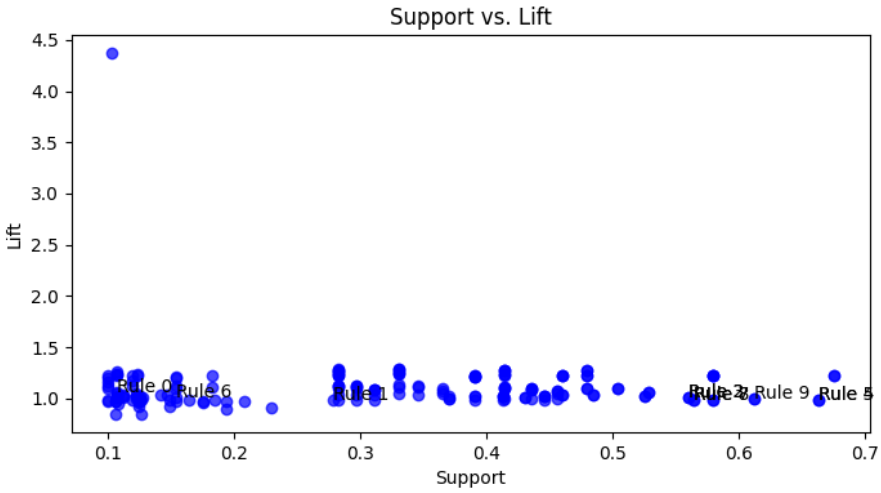

Figure 20: Support vs. Lift

or rare patterns.

These plots collectively highlight the most influential factors in accident severity and guide targeted interventions based on their frequency, reliability, and impact.

In summary, the Apriori algorithm effectively identified meaningful associations between environmental and road-related factors and accident severity. These findings provide actionable insights for traffic management and road safety authorities, aiding in the design of targeted interventions to enhance road safety.

3.5 Accidents in the Region of Kingston upon Hull, Humberside, and the East Riding of Yorkshire

The analysis of accident data for Hull, Humberside, and the East Riding of Yorkshire revealed valuable insights into accident. The dataset consisted of 8,652 accidents, categorized into three severity levels: "Slight" (6,772), "Serious" (1,754), and "Fatal" (126) as shown in Figure 21. Temporal analysis from 2017 to 2020 (Figure 22) showed consistent accident counts in 2017 (2,328), 2018 (2,305), and 2019 (2,310), with a notable decline in 2020 (1,709 accidents), likely due to reduced traffic during the COVID-19 pandemic.

Spatial analysis (Figure 23) revealed a concentration of accidents in urban areas, especially Hull and surrounding regions. Kernel Density Estimation (KDE) analysis (Figure 24) confirmed these high-risk zones, highlighting areas for targeted safety interventions. Accidents were most frequent during peak hours, particularly between 7:00 AM and 9:00 AM, and from 3:00 PM to 6:00 PM, with the highest number recorded at 5:00 PM (782 accidents). Early morning hours (12:00 AM to 5:00 AM) had the fewest accidents, indicating lower traffic volumes during these times.

3.6 Three Policing Areas and Accidents Forecasts

The analysis focused on identifying the three police areas with the highest accident counts from 2017 to 2019 and predicting weekly accident counts for the upcoming year using time series models. The top three areas were the Metropolitan Police, West Midlands, and Kent, with accident counts of 98,092,

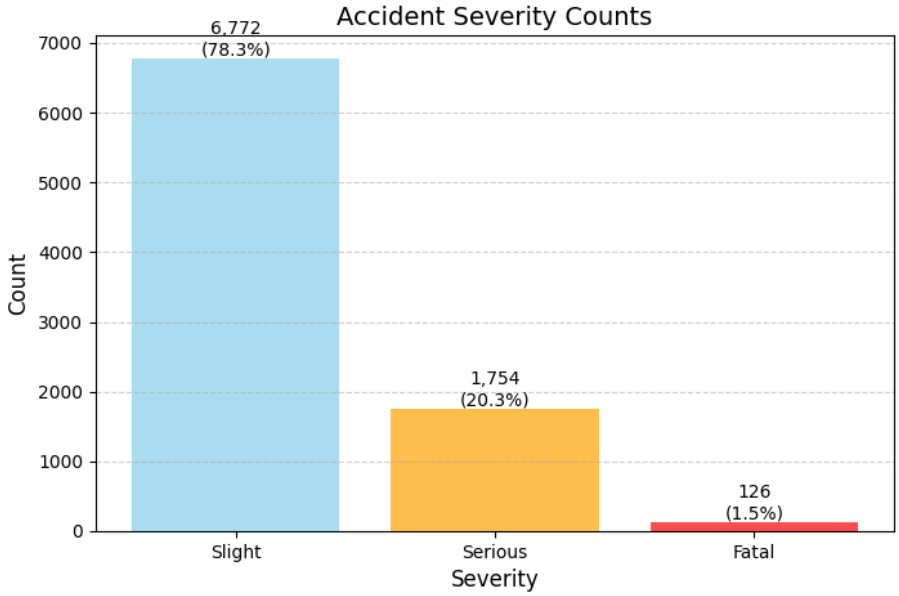

Figure 21: Region Accident severity counts.

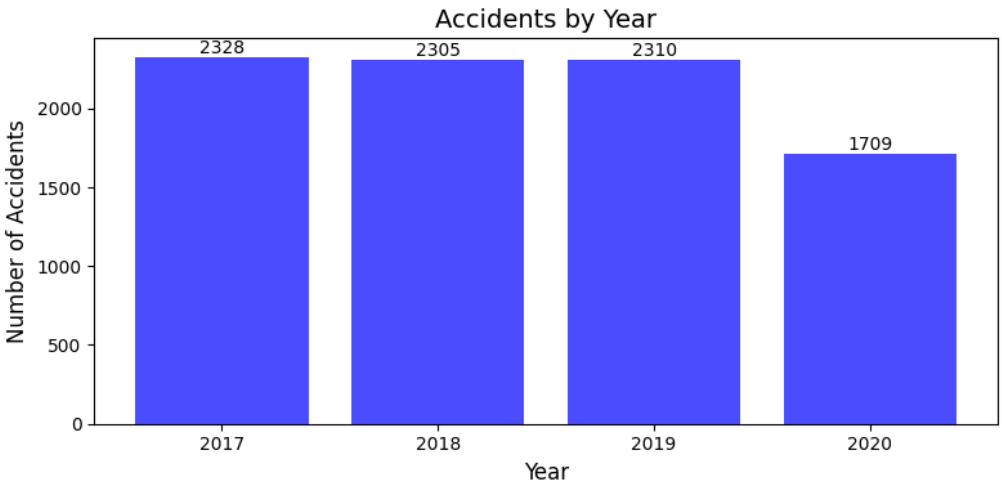

Figure 22: Accidents by Year

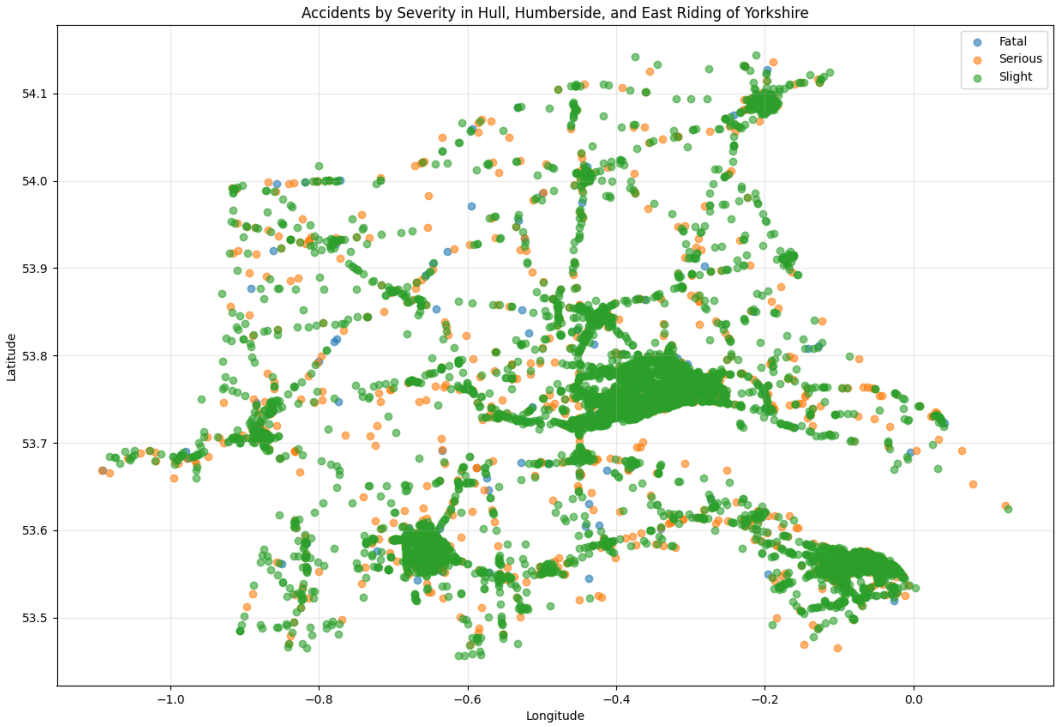

Figure 23: Accidents by Severity in Hull, Humberside, and East Riding of Yorkshire

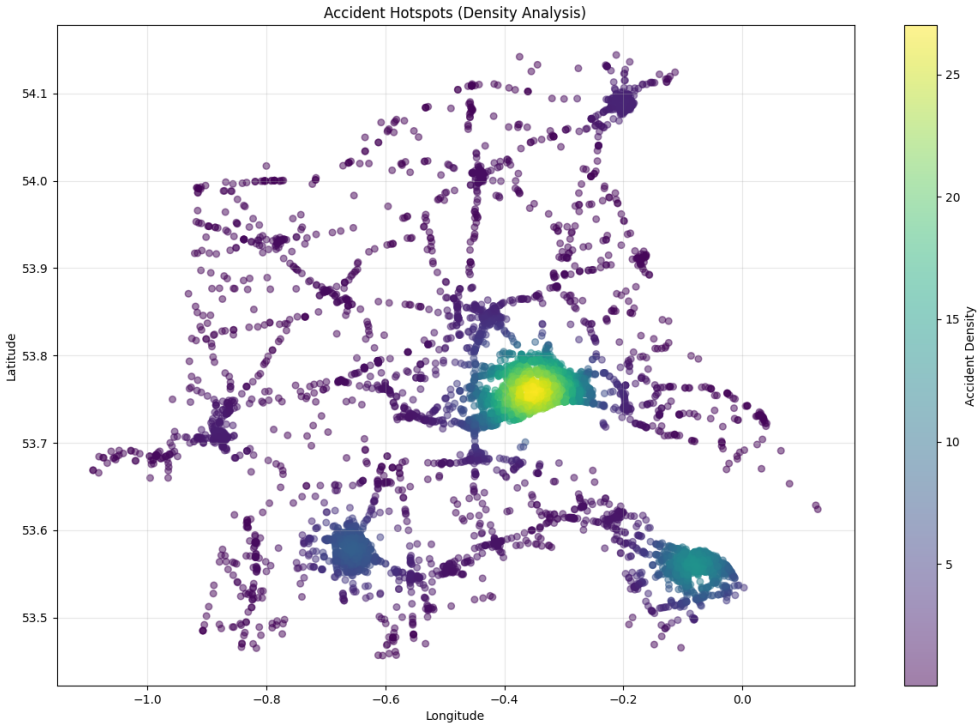

Figure 24: Accident Hotspots (Density Analysis)

20,519, and 16,789, respectively as shown in Figure 25. A SARIMAX model was applied to the weekly

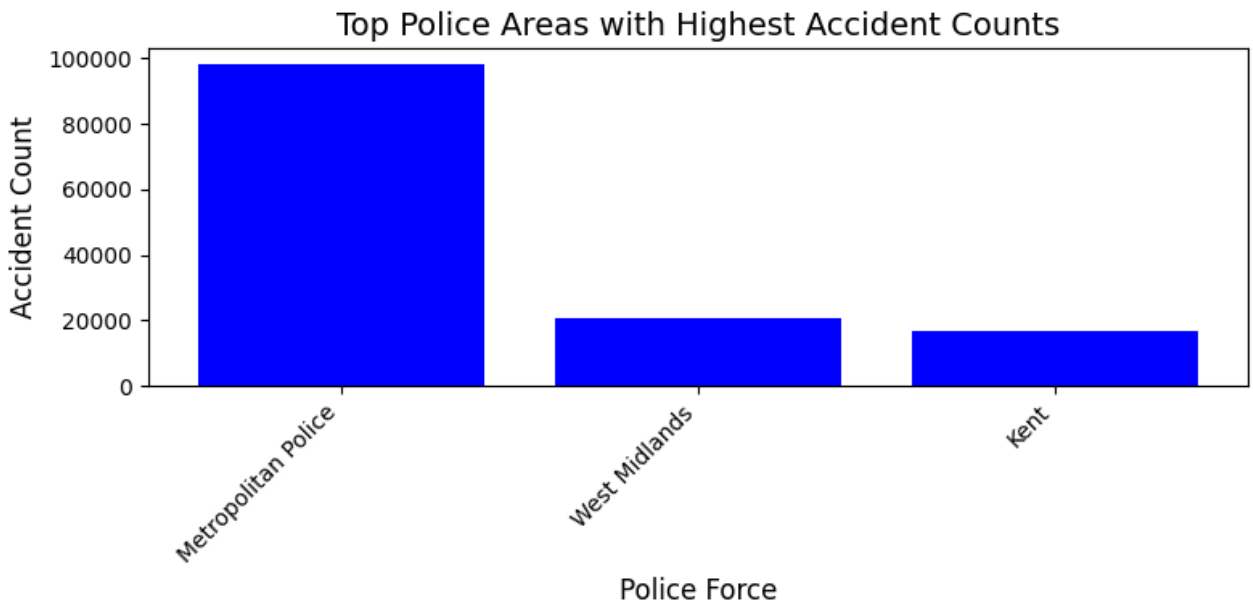

Figure 25: Top Police Areas Accident Counts.png)

accident data from these areas for forecasting trends over the next 52 weeks. The data were grouped into weekly intervals with no missing values in the date column, ensuring the accuracy of the analysis. For the Metropolitan Police, the forecasts (Figure 26) indicated fluctuating accident counts, ranging from a high of 558 during peak travel weeks to a low of 125 during holiday periods. The model showed a gradual decline in accident counts toward the end of the year, reflecting seasonal variations in travel.

The West Midlands forecast (Figure 27) displayed a relatively stable trend, with weekly counts varying between 37 and 132, showing moderate fluctuations. Kent's forecasts (Figure 28) predicted weekly accident counts between 26 and 113, with higher counts in the summer months and lower counts in winter. The model captured seasonal trends, with higher accident frequencies expected during summer and declines in winter. These insights can assist traffic management and safety authorities in preparing for periods of elevated risk and allocating resources effectively.

3.7 Three Local Super Output Areas (LSOAs) of Hull city with Highest accidents and July 2020 Predictions

The analysis identified the top three Local Super Output Areas (LSOAs) in Hull with the highest number of road accidents during the first quarter of 2020. The areas with the most concentrated accident occurrences were E06000010 (603 accidents), E06000011 (495 accidents), and E06000013 (306 accidents) (Figure 29), highlighting their significance for targeted interventions and forecasting. A stationarity analysis using Augmented Dickey Fuller (ADF) shows the series is stationary and as such, does not need differencing. A test time series model (ARIMA(1,0,1)) was used to forecast daily accident occurrences for July 2020 in these LSOAs based on historical data from January to June 2020. For E06000010, the forecast predicted a stable number of daily accidents, ranging from

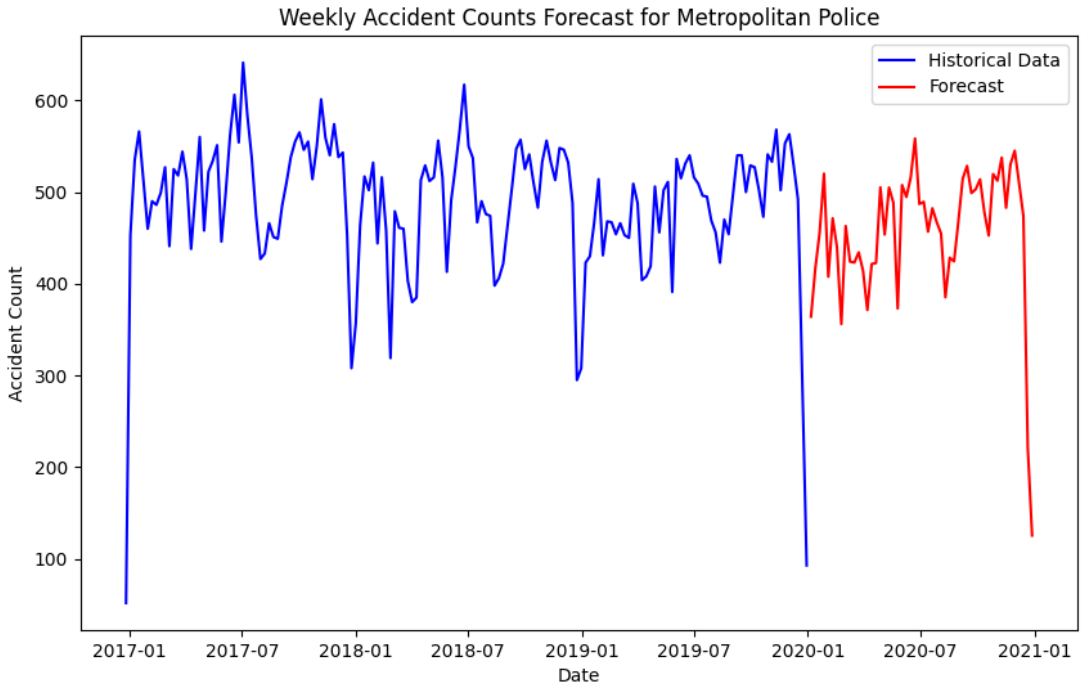

Figure 26: Weekly Accident Counts Forecast Metropolitan Police

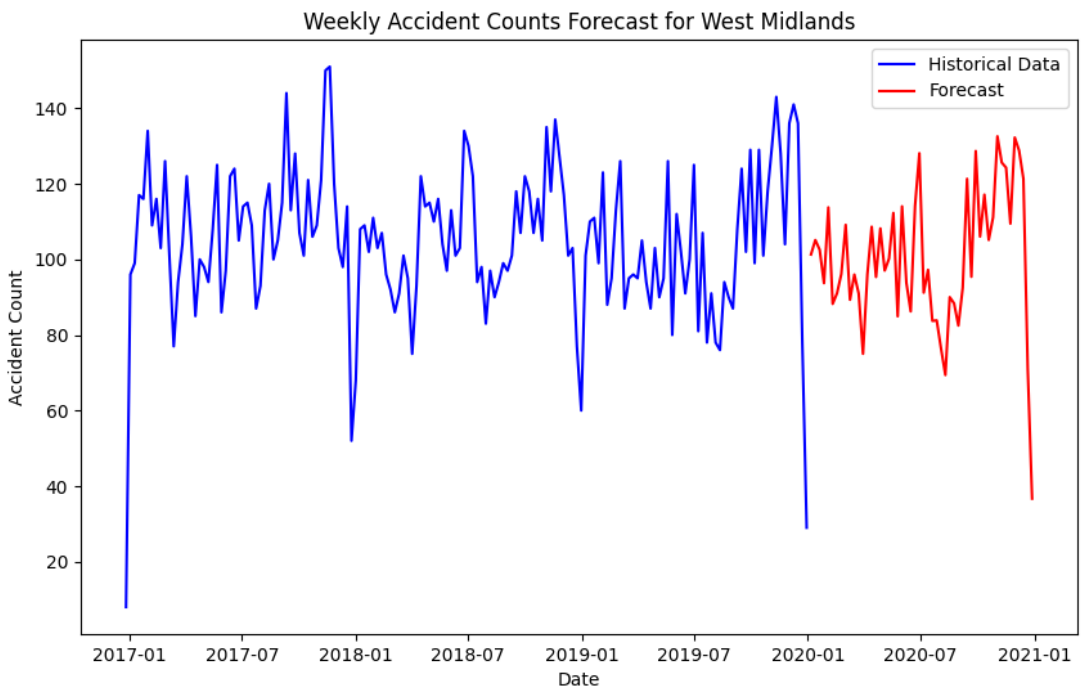

Figure 27: Weekly Accident Counts Forecast West Midlands

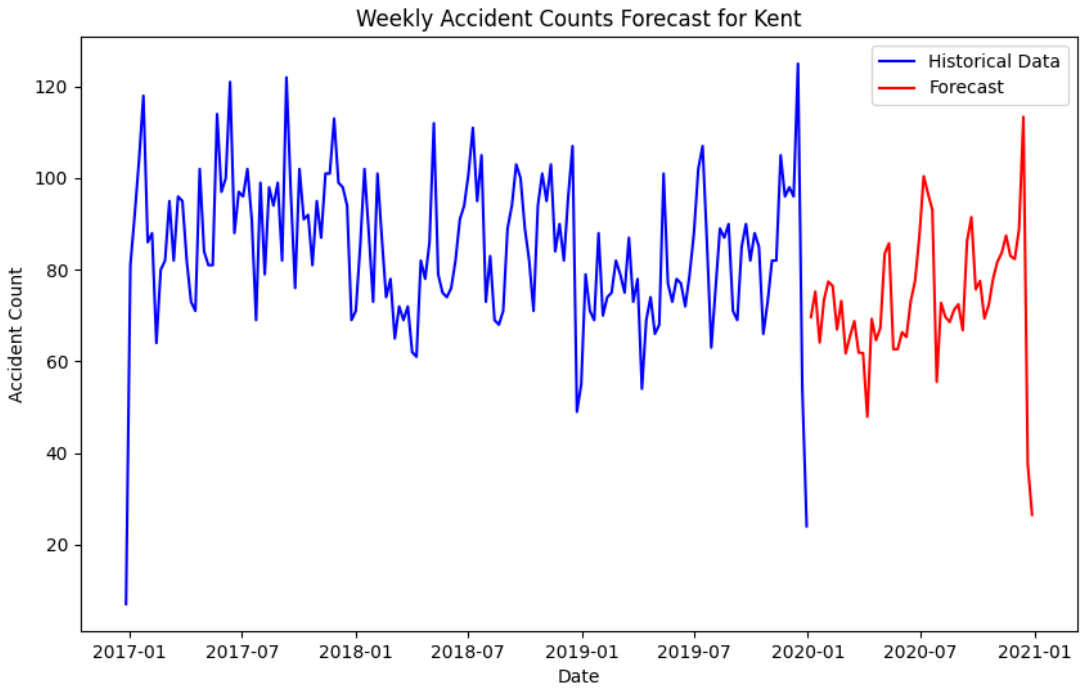

Figure 28: Weekly Accident Counts Forecast Kent

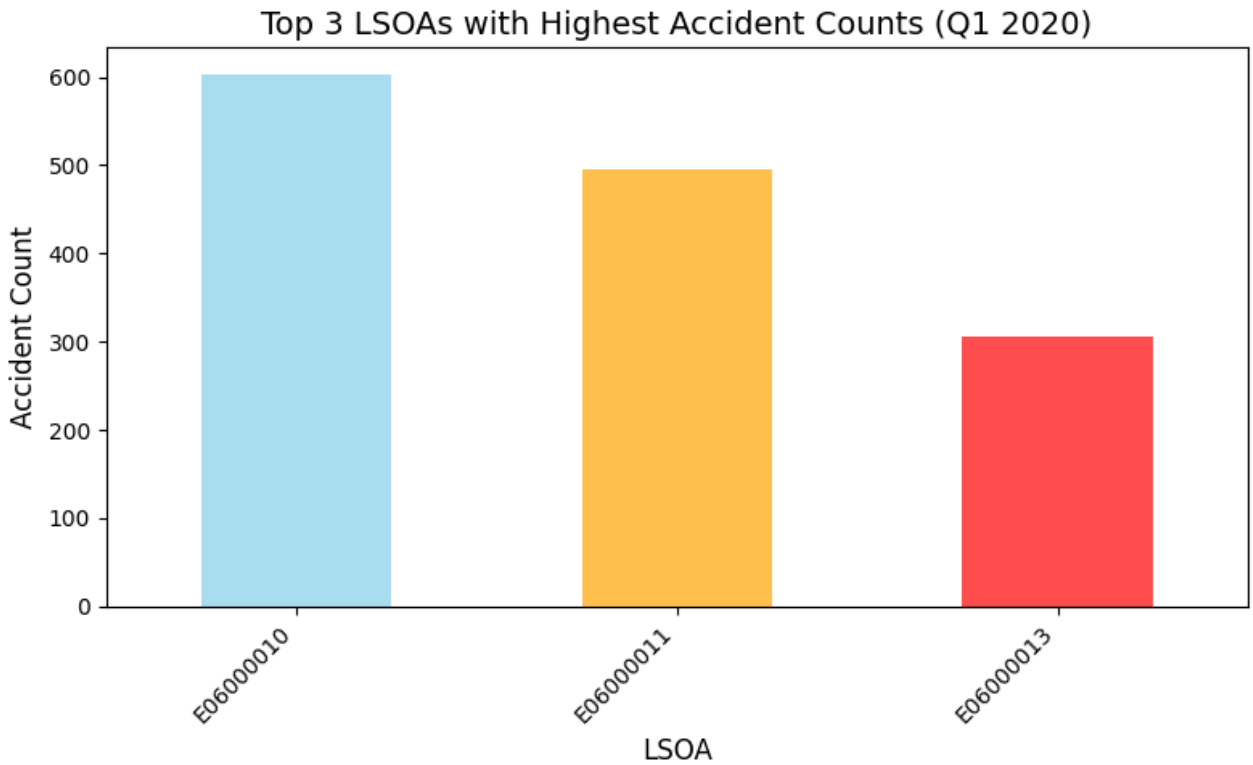

Figure 29: Top 3 LSOAs Accident Counts Q1 2020

approximately 483.28 to 482.07. E06000011 showed a slight decline, with values starting at 519.01 and stabilizing around 495.37. E06000013 indicated a modest decline, with forecasts decreasing from 326.86 to 317.42. These predictions (Figures 30, 31, & 32) provide insights into accident trends, which can assist in planning emergency responses and preventive measures. However, the model's accuracy may be influenced by limitations in the frequency and granularity of the historical data.

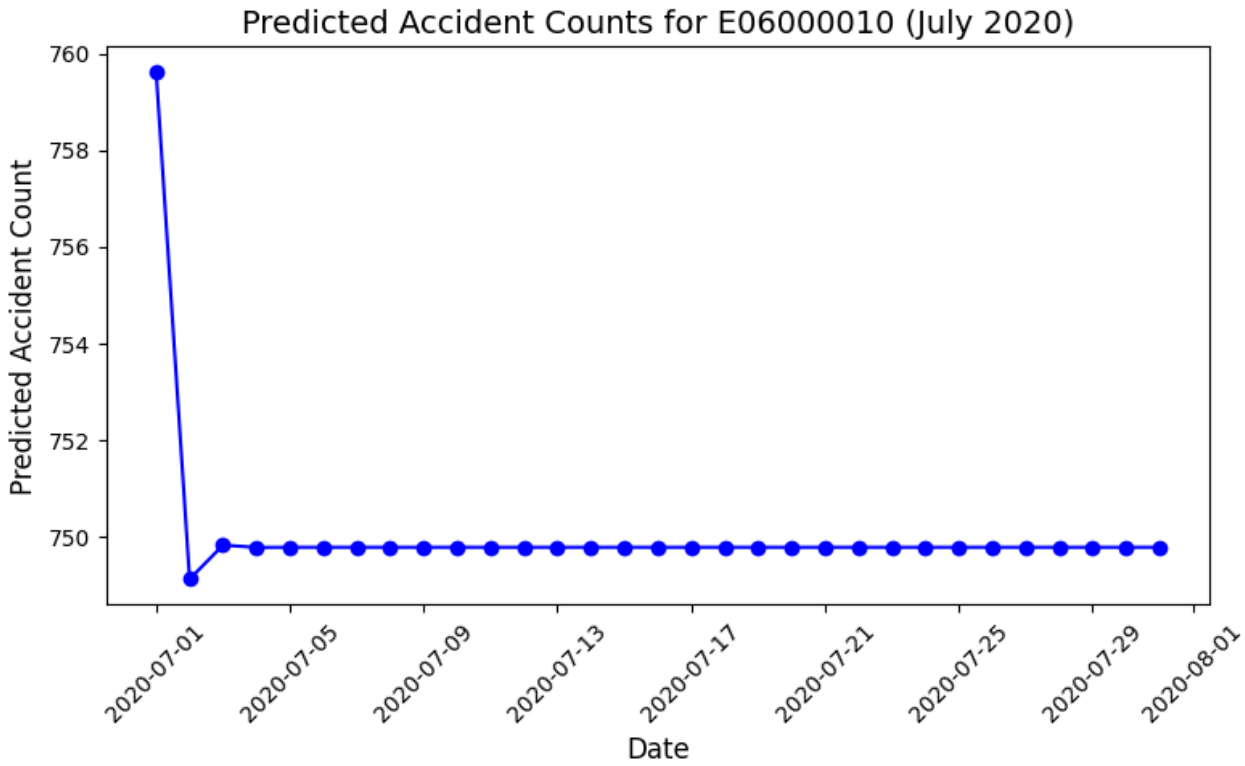

Figure 30: Predicted Accident Counts E06000010 July 2020

3.8 Social Network

A social network was constructed to model interactions between casualties, vehicles, and locations of accidents. The data integration involved merging accident, casualty, and vehicle datasets based on the accident index, forming a unified dataset. Nodes in the network represented three types of entities: casualties, vehicles, and locations, while edges captured the relationships between these entities—such as casualties associated with vehicles, vehicles linked to accident locations, and casualties tied directly to locations. The resulting network had 286 nodes and 844 edges, demonstrating a moderate level of connectivity. The network density was calculated as 0.0207, indicating that only around 2% of all possible connections between nodes were present, characteristic of a sparse network. The average degree (average number of connections per node) was 5.90, suggesting that, on average, each node was connected to almost six other nodes. The visual representation of the network (Figure 33), plotted using a spring layout, highlighted the structure and connections within the graph. This visualization and quantitative analysis provide insights into the interaction dynamics and can serve as a foundation for further investigations, such as identifying central nodes, analyzing connectivity

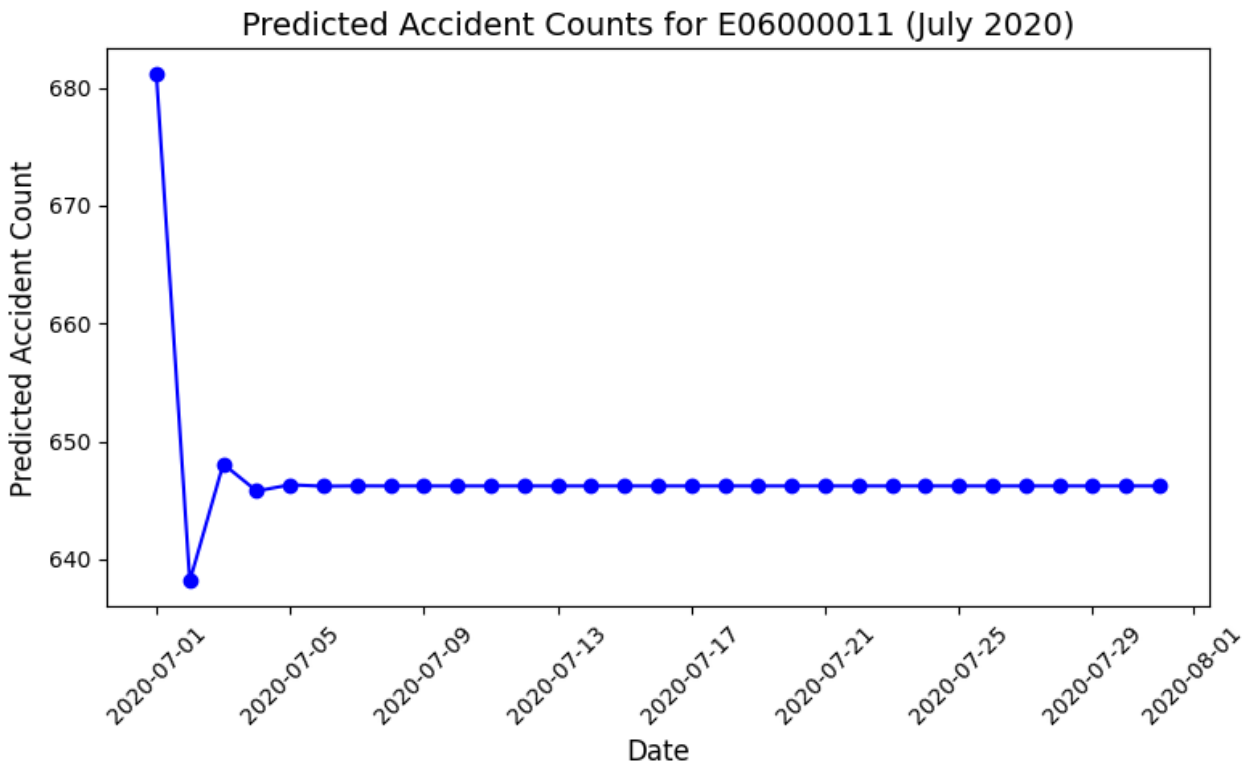

Figure 31: Predicted Accident Counts E06000011 July 2020

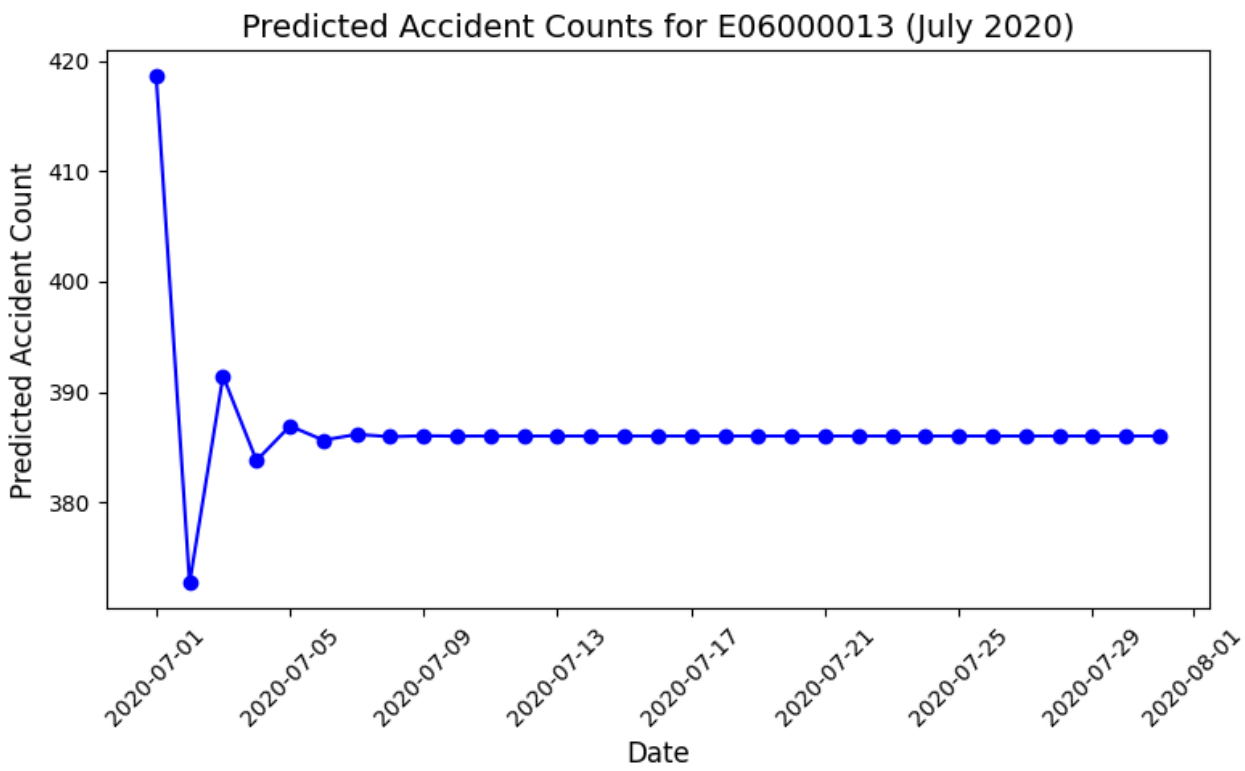

Figure 32: Predicted Accident Counts E06000013 July 2020

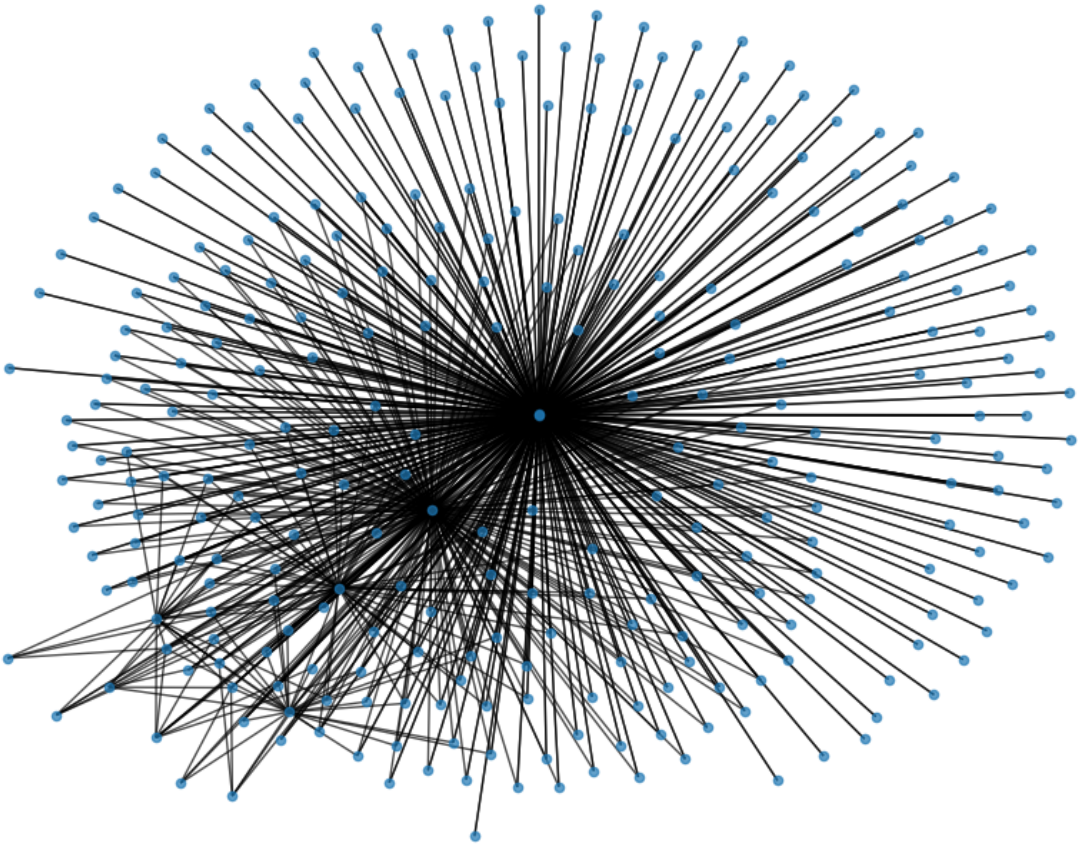

Figure 33: Social Network of Accidents

patterns, or detecting clusters within the network.

3.9 Edge Centrality

The analysis focused on calculating and visualizing the edge betweenness centrality within the constructed social network, which connects casualties, vehicles, and accident locations. Edge betweenness centrality measures the frequency with which an edge lies on the shortest path between pairs of nodes, highlighting the importance of specific edges in maintaining network connectivity. The computed values for edge centrality varied significantly across the network, with values ranging from 0.0001 to 0.0061, as seen in edges such as ('C_1', 'V_1') and ('C_1', 'V_6'), respectively. The visualization of the centrality distribution revealed that most edges had low centrality values, as illustrated by the peak at the lower end of the distribution. This suggests that only a few edges play a critical role in connecting different components of the network, while the majority are less influential. The histogram (Figure 34), enhanced with a KDE (Kernel Density Estimate) curve, provided a clear depiction of the centrality distribution, confirming the sparsity of high-centrality edges. These critical edges may represent key connections between high-impact nodes, such as frequently occurring accident types or densely linked locations. This analysis of edge centrality underscores its importance in identifying pivotal connections within the network, which could guide targeted interventions for accident prevention or response planning. By focusing on high-centrality edges, efforts can be directed to optimize resource allocation or reduce vulnerabilities in the network.

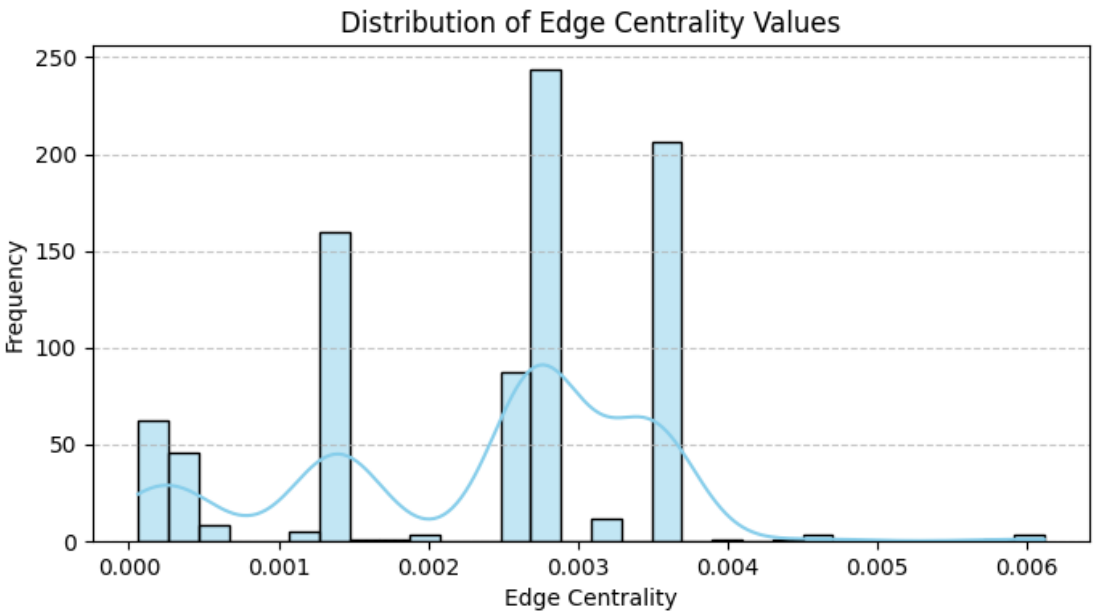

Figure 34: Distribution of Edge Centrality Values

3.10 Two Community Detection Algorithms within the Social Network

The analysis utilized two community detection algorithms, Louvain and Girvan-Newman, to identify clusters within the social network of accidents. The Louvain algorithm identified three clusters: Cluster 0 with 133 nodes, Cluster 1 with 68 nodes, and Cluster 2 with 85 nodes. The distribution of cluster sizes showed considerable variation, highlighting Louvain's capacity to detect diverse communities within the network.

The Girvan-Newman algorithm, on the other hand, identified two clusters. Cluster 0 contained 285 nodes, while Cluster 1 contained just a single node, reflecting the algorithm's tendency to create broader divisions based on edge significance. These algorithms offer different insights into the network's structure. Louvain (Figure 35) reveals smaller, tightly-knit communities, while Girvan-Newman (Figure 36) generates larger, more generalized clusters. The choice of algorithm depends on the specific goals of the analysis, whether focusing on detailed community structures or broader divisions. Both approaches provide valuable insights for targeted interventions or resource allocation based on community structures. A comparison of the two algorithms' community detection results is presented in Figure 37.

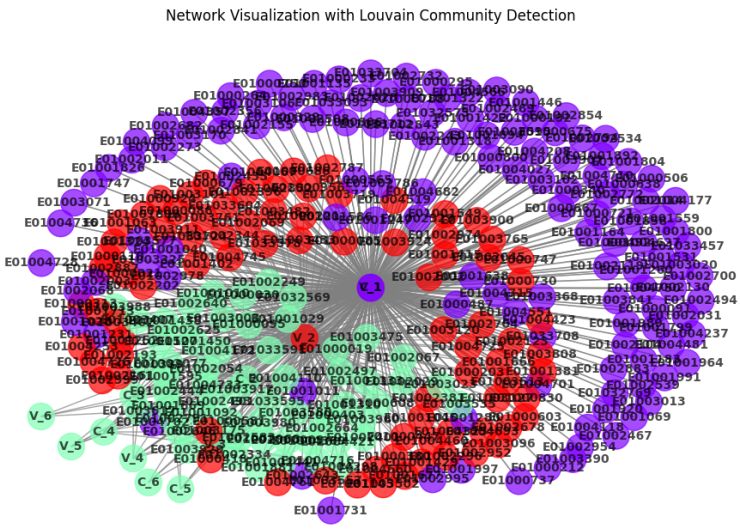

Figure 35: Network Visualization with Louvain Community Detection

4 Discussion of Results

The analysis of road traffic accident data highlights critical areas for intervention in road traffic management, with important implications for reducing accident severity and improving overall road safety. The results reveal that 78.94% of all accidents were classified as slight, indicating that most accidents are relatively minor but still contribute to road safety concerns. However, serious and fatal accidents—comprising 17.33% and 1.20% of the total accidents, respectively—pose a far more significant threat to public safety, emphasizing the need for targeted interventions.

When examining the types of roads involved in accidents, it becomes clear that certain road types are associated with different accident severities. For example, roundabouts, which accounted for 8.63% of the accidents, tend to have a higher proportion of slight accidents. This suggests that, while roundabouts are generally safer, there are still opportunities for improvement in their design, such as clearer signage and better traffic flow management. On the other hand, dual carriageways were associated with a higher number of serious and fatal accidents, making up 24.24% of all serious

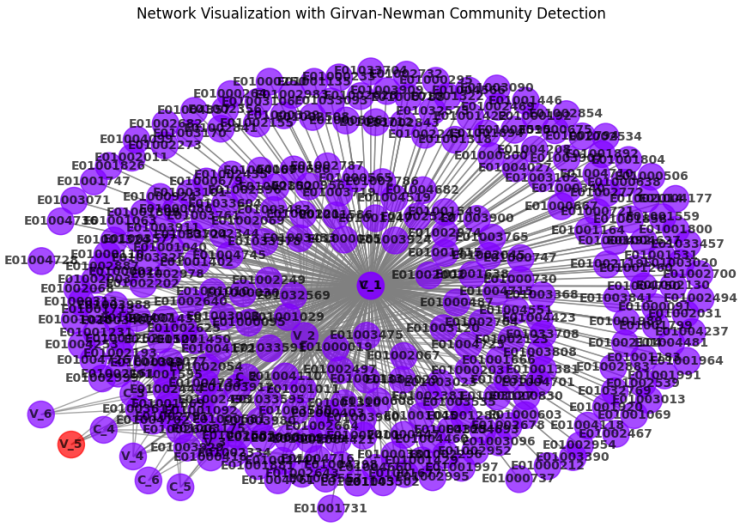

Figure 36: Network Visualization with Girvan-Newman Community Detection

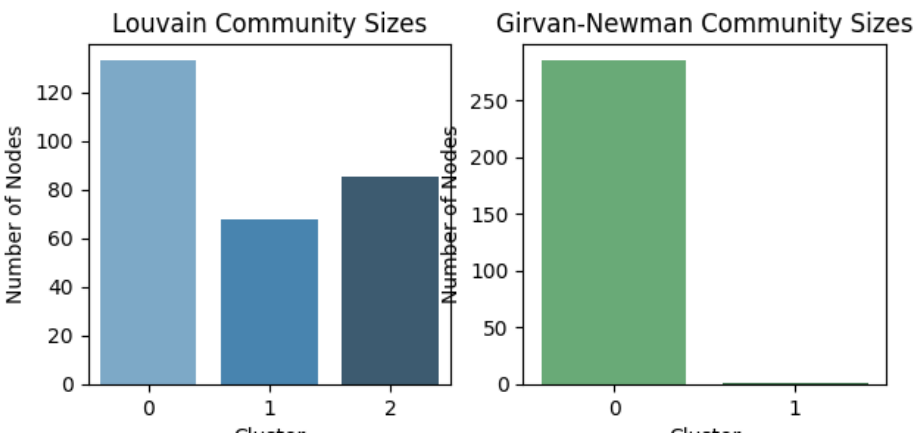

Figure 37: Comparison of Community Detection Results: Louvain vs. Girvan-Newman

accidents. This is consistent with the higher speeds on these roads, which can exacerbate the severity of accidents. Interventions on dual carriageways could include stricter speed enforcement, enhanced barriers, and better road maintenance to improve safety.

Weather conditions also play a crucial role in accident severity. While fine weather accounted for 49.17% of accidents, it is under adverse weather conditions, such as high winds (4.69%) and snow (0.25%), that the severity of accidents increases. Although adverse weather conditions contribute to a smaller proportion of total accidents, their impact is disproportionately large, with higher proportions of serious and fatal accidents during these times. This highlights the need for specific measures during extreme weather events, such as increased public awareness campaigns, weather-sensitive traffic signage, and timely road maintenance to reduce hazards like ice or snow.

Temporal patterns also provide valuable insights into accident trends. The 17:00-18:00 time period accounted for 12.76% of the accidents, correlating with the evening rush hour when traffic volume peaks. These accidents are likely exacerbated by factors such as driver fatigue, congestion, and aggressive driving. Interventions during these times could include enhanced traffic management, dynamic rerouting based on real-time data, and increased law enforcement to discourage risky driving behavior. The analysis also indicates that weekends, especially Saturdays, account for 27.62% of the total accidents. Given that weekends are typically associated with recreational travel, higher alcohol consumption, and increased social activity, targeted interventions like DUI checkpoints and public education campaigns on safe driving during leisure activities could reduce accidents.

Motorcycle accidents follow similar patterns, with accidents peaking during the 17:00-18:00 timeframe and higher rates on weekends, particularly with 50cc and under motorcycles. Smaller motorcycles, often used for recreational purposes, tend to have higher accident rates on weekends, suggesting that improving helmet laws, road safety education, and infrastructure for motorcyclists could help mitigate these risks. Larger motorcycles, on the other hand, which are typically used for commuting, experience higher accident rates during weekdays, necessitating more targeted interventions like speed enforcement, road condition monitoring, and defensive driving campaigns.

Pedestrian accidents also show a clear temporal pattern, with the highest incidence occurring during the 8:00-9:00 and 16:00-17:00 time frames, coinciding with peak foot traffic during school runs, work commutes, and after-school activities. In total, pedestrian accidents accounted for 11.16% of all accidents in the dataset. These accidents could be mitigated by improving pedestrian infrastructure, such as better crossings, clearer signage, and increased pedestrian safety enforcement, especially in high-traffic areas near schools or shopping districts.

The Apriori algorithm provided further insights into the relationships between accident severity and various contributing factors. For instance, accidents occurring on dry roads were predominantly classified as slight (83.72%), suggesting that while dry conditions do not contribute significantly to severe accidents, road conditions still play a role in the outcome. On dual carriageways, which were associated with 24.24% of all serious accidents, the relationship between road type and severity is clear, reinforcing the need for safety measures like speed regulation, barriers, and traffic flow improvements on these high-risk roads.

In terms of predictive modeling, time series forecasts indicate that accident rates are expected to rise during peak travel periods, such as the summer months, public holidays, and weekends, where higher volumes of traffic and riskier driving behaviors are prevalent. By forecasting these trends, traffic authorities can allocate resources more effectively, ensuring that police presence, emergency services, and traffic management systems are enhanced during these high-risk times.

A deeper analysis using social network analysis (SNA) helped to further reveal hidden patterns in the road traffic accident data. A social network was constructed to model interactions between casualties, vehicles, and accident locations. The resulting network, which included 286 nodes and 844 edges, demonstrated a moderate level of connectivity, with an average degree of 5.90, meaning each node was connected to nearly six others. The network density of 0.0207 indicates that only about 2%

of possible connections were present, which is typical for a sparse network. The centrality analysis of this network showed that only a few edges were critical in maintaining network connectivity, with edge betweenness centrality values ranging from 0.0001 to 0.0061. Most edges exhibited low centrality, but critical connections between certain accident types, casualties, vehicles, and locations could play a vital role in improving traffic safety management.

The Louvain and Girvan-Newman community detection algorithms helped uncover the structure of these accident networks. The Louvain method identified three clusters with varying sizes: Cluster 0 contained 133 nodes, Cluster 1 had 68 nodes, and Cluster 2 had 85 nodes. These clusters likely represent distinct groups of accidents that could be targeted with tailored interventions. The Girvan-Newman method, on the other hand, detected only two large clusters, with one containing 285 nodes and the other only one, showing a sharper division based on edge betweenness. This difference in clustering highlights the complementary nature of these methods, where Louvain can capture more granular community structures, while Girvan-Newman provides a broader division based on the most significant connections.

By focusing on high-centrality edges and the communities identified, targeted interventions can be designed to optimize accident response and prevention efforts. For instance, areas identified as accident hotspots in the network could receive enhanced traffic management measures or more frequent patrols. Communities with frequent interactions between certain casualties and vehicles could benefit from education campaigns that specifically address those risk factors. In conclusion, the findings highlight the need for targeted, data-driven interventions to address the most significant risk factors contributing to road traffic accidents. With nearly 79% of accidents categorized as slight, the focus should be on reducing the severity of those that occur. For more serious and fatal accidents, which account for over 18% of total accidents, efforts should concentrate on high-risk areas such as dual carriageways, adverse weather conditions, peak times like rush hour, and weekends. By leveraging predictive analytics, improving infrastructure, increasing enforcement, and applying insights from social network analysis to understand the dynamics of accident-prone communities, authorities can develop more effective strategies that ultimately reduce both the frequency and severity of road traffic accidents, leading to safer roads for all users.

5 Conclusion

This study has provided a comprehensive analysis of road traffic accident data, identifying key factors that contribute to accident severity. By examining accident characteristics such as road types, weather conditions, temporal patterns, and contributing factors like motorcyclist and pedestrian accidents, targeted interventions can be developed to address the most significant risks. Predictive modeling and social network analysis further emphasized the role of data-driven approaches in uncovering hidden patterns within accident data, enabling more effective traffic management strategies.

The findings underscore the need for tailored interventions to improve road safety, focusing on high-risk areas such as dual carriageways, adverse weather conditions, and peak traffic times. The results also highlight the potential of predictive analytics and community detection algorithms in identifying accident hotspots and vulnerable communities, paving the way for enhanced enforcement, better infrastructure, and more efficient resource allocation. Ultimately, the goal is to reduce the frequency and severity of road traffic accidents, contributing to safer roads for all users.

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